

JOAO BASSO

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EDUCATION

University of California, Berkeley, USA

August 2022 - present

Ph.D. in Mathematics.

Advisors: Profs. Lin Lin & Nikhil Srivastava.

Tufts University, Medford, USA

September 2016 - May 2020

Bachelor of Science, Summa Cum Laude, 3.91/4.0 GPA: Mathematics, Physics and Computer Science triple-major.

High Honors in Thesis: “Coordinate-free Tensor Analysis”.

Activities: Tufts Symphony Orchestra (concertmaster), Tufts Chamber Orchestra (associate concertmaster), Tufts Youth Philharmonic (mentor).

PUBLICATIONS AND PRE-PRINTS

21. *J. Basso, T. Bergamaschi, L. Lin, M. Ragone, K. Stubbs.* Spectral Gap of the Davies Generator for the Mean-Field Heisenberg Model, *arXiv:2607.21798* (2026).

20. *J. Basso, S. Ganguly, A. Sinclair, N. Srivastava, Z. Stier, T.-D. Vuong.* On quantum to classical comparison for Davies generators, *QIP 2026*.

19. *Z. Chen, J. Basso, Z. Ding, L. Lin.* Quantum Replica Exchange, *arXiv:2510.07291* (2025).

18. *J. Basso, C. Chen, A. Dalzell.* Optimizing random local Hamiltonians by dissipation, *QIP 2025*.

17. *L. Zhou, J. Basso, S. Mei.* Statistical Estimation in the Spiked Tensor Model via the Quantum Approximate Optimization Algorithm, *NeurIPS 2024*.

16. *S. Sureshbabu, D. Herman, R. Shaydulin, J. Basso, S. Chakrabarti, Y. Sun, M. Pistoia.* Parameter Setting in Quantum Approximate Optimization of Weighted Problems, *Quantum*. (2023)

15. *D. Berry, Y. Su, C. Gyurik, R. King, J. Basso, A. Barba, A. Rajput, N. Wiebe, V. Dunjko, R. Babush.* Analyzing Prospects for Quantum Advantage in Topological Data Analysis, *QIP 2022, PRX Quantum*.

14. *Google Quantum AI and Collaborators.* Suppressing quantum errors by scaling a surface code logical qubit, *Nature*. (2022)

13. *Google Quantum AI and Collaborators.* Formation of robust bound states of interacting photons, *Nature*. (2022)

12. *Google Quantum AI and Collaborators.* Noise-resilient Majorana Edge Modes on a Chain of Superconducting Qubits, *Science*. (2022)

11. *J. Basso, D. Gamarnik, S. Mei, L. Zhou.* Performance and limitations of the QAOA at constant levels on large sparse hypergraphs and spin glass models, *FOCS 2022*.

10. *J. Basso, E. Farhi, K. Marwaha, B. Villalonga, L. Zhou.* The Quantum Approximate Optimization Algorithm at High Depth for MaxCut on Large-Girth Regular Graphs and the Sherrington-Kirkpatrick Model, *TQC 2022, Outstanding Paper Prize*.

9. *Google Quantum AI and Collaborators.* Time-Crystalline Eigenstate Order on a Quantum Processor, *Nature*. (2021)

8. *Google Quantum AI and Collaborators.* Information scrambling in quantum circuits, *Science*. (2021)

7. *Google Quantum AI and Collaborators.* Realizing topologically ordered states on a quantum processor,

Science. (2021)

6. J. Basso, L. Tu. Basis-free analysis of singular tuples and eigenpairs of tensors, *arXiv:2012.07313*. (2020)

5. M. Mosca, J. Basso, S. Verschoor. On speeding up factoring with quantum SAT solvers, *Nature Scientific Reports*. (2020)

4. J. Basso, I. Yurchenko, M. Wiens, C. Staii. Neuron dynamics on directional surfaces, *Soft Matter*. (2019)

3. I. Yurchenko, J. Basso, V. Syrotenko, C. Staii. Anomalous diffusion for neuronal growth on surfaces with controlled geometries, *PLoS One*. (2019)

2. J. Basso, I. Yurchenko, M. Simon, D. Rizzo, C. Staii. Role of geometrical cues in neuronal growth, *Physical Review E*. (2019)

1. J. Basso, M. Simon, C. Staii. Neuronal dynamics on patterned substrates measured by fluorescence microscopy, *MRS Communications*. (2018)

WORK EXPERIENCE

JPMorgan Chase & Co., New York, NY, USA
Summer Associate

June 2024 - August 2024

- Quantum computing research under the Global Technologies Applied Research division.

NASA Quantum AI Lab, Mountain View, CA, USA
Research Assistant, Quantum

August 2023 - May 2024

- Work on quantum algorithms through the Universities Space Research Association (USRA). Divided into two periods, August 2023 - January 2024 and January 2024 - May 2024, the latter of which I worked remotely.

Amazon.com Inc., Pasadena, CA, USA
Research Scientist Intern

May 2023 - August 2023

- Worked on the development and analysis of quantum algorithms at the AWS Center for Quantum Computing.

Google Research - Quantum AI, Venice, CA, USA
AI Resident

August 2020 - August 2022

- Worked on the development and analysis of quantum algorithms.

TEACHING EXPERIENCE

Mount Tamalpais College, San Quentin, CA, USA
Instructor

2025

- Volunteered as an instructor for the calculus class. Mount Tamalpais, formerly known as the Prison University Project, is an accredited college for inmates in the San Quentin prison.

University of California, Berkeley, CA, USA
Graduate Student Instructor

2022 - present

- **Calculus I** (Fall 2022), **Calculus II** (Fall 2024).

Tufts University, Medford, MA, USA
Teaching Assistant

2017 - 2020

- **Complex Variables** (Spring '19), **Algorithms** (Summer '18), **Linear Algebra** (Fall '19), **Discrete Mathematics** (Fall '17, Spring '18, Fall '18), **Intro to Physics 1** (Fall '17, Fall '18), **Intro to Physics 2** (Spring '18, Spring '19), **General Physics 11** (Fall '17, Fall '19), **General Physics 12** (Fall '17, Spring '20), **Portuguese** (Spring '17, Fall '17, Spring '18).

HONORS AND AWARDS

Jane Street Graduate Research Fellowship Award: Honorable mention. (2024)

Outstanding Paper Prize: 17th Conference on the Theory of Quantum Computation, Communication and Cryptography. (TQC 2022)

2020 Senior Award: Awarded to six to 12 graduating members of the senior class for academic achievement, participation in campus and community activities, and leadership. (Tufts University Alumni Association, April 2020).

Benjamin G. Brown Scholarship: Awarded to Tufts seniors who have shown promise in scientific research. (Tufts University, April 2020)

Tufts Career Center Internship Grant: Received for a summer research internship at Boston University. (Summer 2018)

Howard Sample Prize Scholarship in Physics: Awarded for outstanding performance in General Physics I and II, calculus-based. (Tufts University, March 2018)

Portuguese Conversation Group Leader: Award of Excellence. (Tufts University, May 2017)

Brazilian Physics Olympiad: 1 Silver ('13), 1 Bronze ('12) and 1 Honorable Mention ('14) medals.

Physics Olympiad of Sao Paulo: 2 Bronze ('14, '10) medals.

Brazilian Astronomy and Astronautics Olympiad: 5 Gold ('08, '10, '12, '14, '15) and 3 Silver ('09, '11, '13) medals.

FIRST LEGO League: 1 Robot Design ('10) and 2 Core Values ('11, '13) trophies.

SERVICE

Paper refereeing: PRL; PRA; PRX Quantum; FOCS 2026; TQC 2026; QIP 2026; TQC 2025; QIP 2025; SODA 2023; TQC 2022; Nature Scientific Data; IEEE Transactions on Quantum Engineering; Quantum Journal; Quantum Science and Technology.

Session chair for “Understanding and Mitigating Decoherence in Superconducting Qubits” at APS March Meeting, 2023.

LANGUAGES

English (fluent); German (intermediate); Portuguese (native).